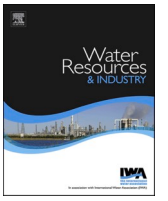
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Municipal reclaimed water as makeup water for cooling systems: Water efficiency, biohazards, and reliability

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ABSTRACT

Municipal reclaimed water is increasingly being considered as a sustainable source of makeup water for cooling towers. However, many oil and gas facilities in the water scarce middle eastern regions still depend on non-renewable groundwater for operating cooling towers. Successful application of reclaimed water in a dry desert climate with very high temperature requires a critical understanding of several factors, including water efficiency, biohazards, equipment reliability, and economics. In particular, biohazard assessment, due to bacteria and virus contamination in hot weather, should be assessed to understand the human risks of utilizing reclaimed water. This study investigated the applicability of municipal reclaimed water as an alternative to groundwater for an industrial cooling system at an oil and gas facility in Saudi Arabia. Treated wastewater effluent from an activated sludge biological treatment process was utilized, as make up water for a 4.2 MW cooling tower. The results of this study showed that water efficiency can be increased using municipal reclaimed water instead of groundwater by increasing the cycle of concentration. The bacterial and enteric virus study indicated that the disinfection and biocide treatment was effective in controlling biological growth and minimizing associated risks to human health. The use of reclaimed water was economically viable with 27% reduction in water consumption.

1. Introduction

Oil and gas companies operating in severely water scarce middle east region is constantly looking for alternatives for groundwater. Water related risk for an industry can be physical, regulatory, or reputational [1]. Physical risk occurs when regional water supplies are not always available at the quantities or qualities needed to adequately support the myriad of community water dependent needs. Water availability for industrial use is impacted by many factors, including location of existing freshwater resources, water management practices, competing interests for water supplies, development related pollution, and climate change. Water reuse and recycling investments is a sustainable solution for regional water scarcity concerns and/or wastewater discharge limitations [2]. Effective water reuse applications can deliver environmental, economic and social benefits to the area it operates [3]. In addition, water reuse plays an important role in shifting from linear to circular business models and to make water-smart circular economy. The

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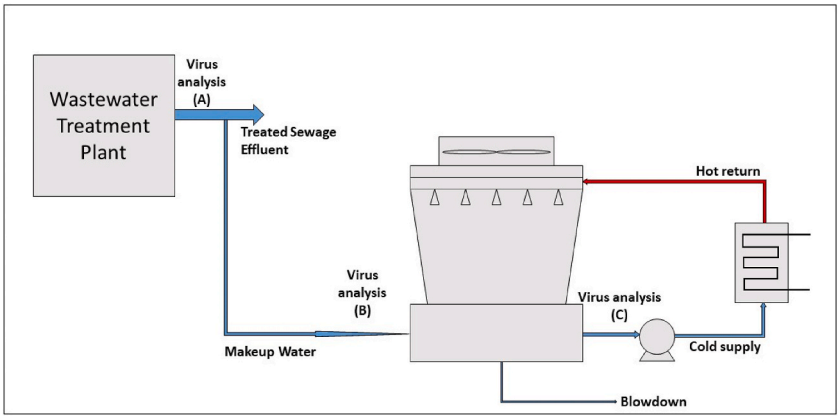


Fig. 1. Conceptual diagram of an evaporative cooling system, showing virus sampling locations for the study.

use of alternative water supplies and water reuse initiatives are increasingly being considered [4–6]. Treated municipal effluent has already been considered by the power sector to potentially offset water withdrawal and consumption for beneficial uses in many parts of the world [4,7–10]. A wide-scale application of municipal reclaimed water for industrial application could assist in offsetting groundwater withdrawal in the middle east region.

Open-recirculating evaporative cooling towers are the most commonly used cooling technology for commercial air conditioning. In open-recirculating systems, water is withdrawn from the cooling tower's basin to the process/equipment that requires cooling and returned back to the top of the cooling tower. Continuous evaporation of the cooling water results in losses of the water and in increase in total dissolved solid concentration of recirculating water. This problem is addressed by continuous water makeup to and blowdown from the basin. A conceptual schematic of an evaporative cooling system is presented in Fig. 1. Normal blowdown adds to the evaporation losses another (0.1–1.0%) of the recirculation flow and the normal water demand (makeup) is only 0.9–1.7% of the recirculation flow but can be as high as 5% if the blowdown is not properly controlled. Drift accounts for a very small portion of uncontrolled losses in the tower. Most towers are installed with drift eliminators that limit drift to 0.005% of system water volume. Water evaporation within the evaporative cooling tower accounts for the majority of the heat rejected, typically 75–80% of the total heat removed. Typically, cooling towers can consume 20 to 30% of a facility's total water use. Thus, to minimize the dependence on freshwater sources for cooling tower operations, the use of alternative water supplies and water reuse initiatives are increasingly being considered by industries [11–16].

Previous studies reported that municipal reclaimed water as cooling system make-up water may pose challenges with respect to enhanced biofouling, corrosion, and scaling [7]. The water quality also determines the cycle of concentration (also known as concentration ratio) of the operation, indicating how efficiently water is recirculated before being lost to blowdown. An increase in the cycle of concentration (COC) reduces water consumption by reducing makeup water volume, and wastewater generation (blowdown). However, wide-scale application of reclaimed water in the oil and gas industry, particularly in the middle eastern region is limited. Thus, a critical understanding of the water quality and operational performance of a cooling system using reclaimed water as makeup water through long-term pilot testing in a hot and humid middle eastern region is a contemporary research need.

One of the challenges of using municipal reclaimed water is biohazards concern in hot environment. When water temperatures are in the range of 20°–50 °C (68°-122 °F), cooling towers have the potential for microbiological growth. Potential proliferation of bacterial and virus must be controlled for successful implementation. The concentration of bacteria and viruses in raw wastewater covers a wide range [17]. While primary and secondary treatment processes can reduce virus concentrations, they are not specifically designed for this purpose, and therefore tertiary treatments such as filtration are often integrated to provide a multiple barrier approach. Cooling towers have been implicated in many outbreaks of Legionnaire's disease worldwide [18,19]. Bacteria can multiply under favorable nutrient rich condition and cause pneumonia, particularly in immune-compromised individuals, through ingestion of contaminated water or inhalation of contaminated aerosols with dispersion extended over a distance of at least 6 km [23]. Human exposure through bio-aerosols relate primarily to 'drift' from the integral drift eliminators, 'windage' directly from the cooling tower basin and fugitive exposures arising from any structural leakage [20]. The risk to populations from aerosols infected with viruses arising from a cooling tower system is primarily theoretical, not readily borne out by published research on outbreaks. Viruses does not replicate outside of the host and their environmental fate will be dependent on a range of environmental factors including temperature, humidity and sunlight. However, it is recognized that enteric viruses, including Rotavirus and Adenovirus can present a risk to human health in certain situations, such as irrigation of landscapes in un-restricted areas, and research has shown that use of a primary biocide such as sodium hypochlorite, which is the most popular disinfection method may not in itself be able to effectively remove viruses from treated effluent. Thus, understanding of the bacteria and virus in the municipal reclaimed water application and effective disinfection system is critical to protect the public health.

To-date, little to no studies have been documented to provide a critical understanding of the effectiveness water treatment for controlling biohazards of utilizing reclaimed water in dry dessert environment with high temperature. The novelty of this paper included: (i) this research represents first of its kind in the Middle Eastern region, which has established a platform for viral detection in

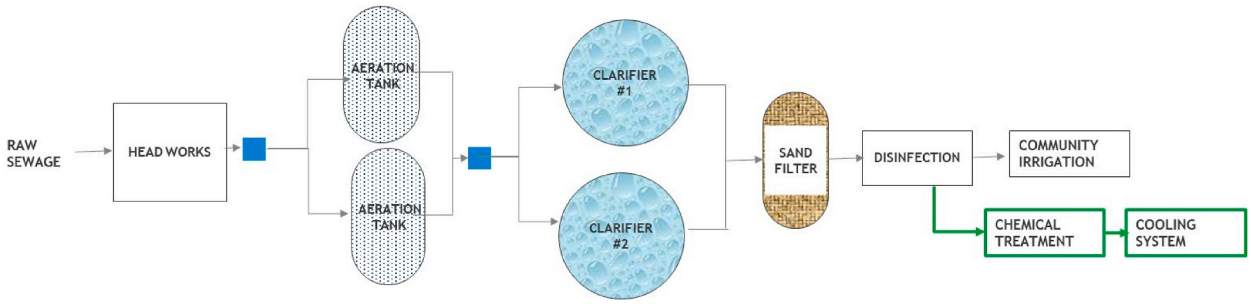


Fig. 2. Conceptual Process Flow diagram of the wastewater treatment plant (green boxes are shown to locate the study). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1
Water quality of Raw Groundwater and Reclaimed Water of the study.

Constituents	Concentration (mg/L)			
	Raw Groundwater	Reclaimed Water		
		Min	Ave	Max
Sodium	680	90	379	840
Potassium	41	5	25	57
Calcium	233	87	94	99
Magnesium	97	31	41	74
Strontium	5	0.08	2	4.5
Chloride	1170	400	565	1233
Sulfate	558	219	566	1110
Bicarbonate	226	50	70	131
Carbonate	0	0	0	0
TDS	3040	1070	1500	1800
Phosphate	–	7.2	8.15	9.8
Nitrate	–	11	12	13
Conductivity (μS/cm)	4770	1233	2370	5480
pH	7.82	6.8	7.4	7.9
Alkalinity (as $CaCO_3$)	186	40	57	107
Hardness (as $CaCO_3$)	981			
Ammonia	–	<0.5	0.2	0.7
TSS	–	–	<5	–
BOD	–	–	<5	–
TOC	–	–	5.12	–

Note: "-" means the data is not available.

reclaimed water samples by the latest and advanced molecular diagnostic method, (ii) demonstrates that reclaim water without any expensive advanced water treatment (e.g., demineralization/reverse osmosis) can replace freshwater/groundwater consumption for cooling make up water in the dry dessert environment regions where groundwater is a nonrenewable water supply; and (iii) provides a conceptual decision matrix on the application of reclaimed water for cooling systems. The objective of the paper is to present the application of municipal reclaimed water as make-up water for cooling system based on water efficiency, biohazard, equipment reliability, and economics.

2. Materials and methodology

2.1. Wastewater treatment plant description

The municipal reclaimed water was generated from an activated sludge wastewater treatment plant with a capacity to treat an average daily flow of 13,400 m^3/day at an oil and gas facility in Saudi Arabia. A conceptual diagram of the plant is presented in Fig. 2. The headworks provides preliminary treatment using bar racks to remove debris (rags, rock material, wooden material) and grit chamber that allows settlement of heavier material. Preliminary treated WW from headworks flows to splitter boxes where it is equally split for the aeration tanks. Secondary wastewater treatment is achieved in the aeration tanks. Surface aerators in aeration tank are operated to maintain the required level of dissolved oxygen (2.0 mg/L) for microbiological degradation of organic matter. The DO level is monitored every day and surface aerator speed is adjusted manually as required. Daily samples are collected to validate the aeration tank process performance for dissolved oxygen, pH, temperature.

Clarifier effluent water from the activated sludge process is continuously injected with a coagulant (Alum) to enhance the sand

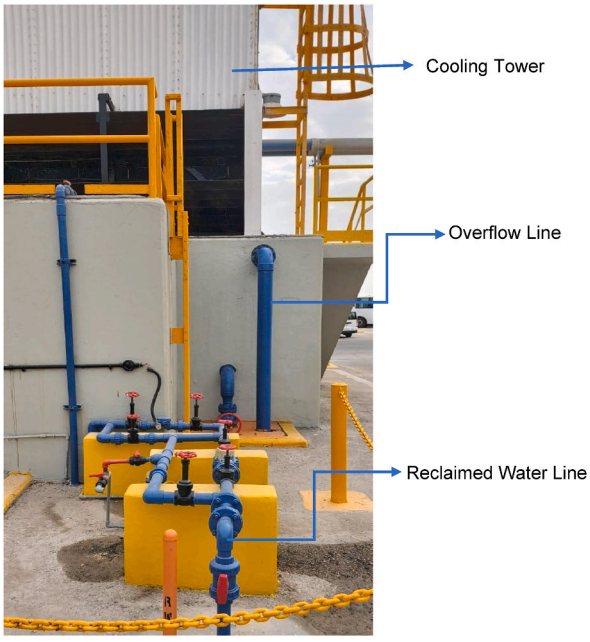


Fig. 3. Cooling tower and reclaimed water connection line used in the study.

Table 2
Chemical analysis parameters and analytical methods.

Parameters	Methods
Sodium	ASTMD1976
Potassium	ASTMD1976
Calcium	ASTMD1976
Magnesium	ASTMD1976
Strontium	ASTMD1976
Chloride	ASTMD512
Sulfate	ASTMD516
Bicarbonate	ASTMD3875
Carbonate	ASTMD3875
Hydroxide	ASTMD3875
TDS	ASTMD1125
Phosphate total	ASTMD515
Nitrate	ASTMD3867
Conductivity (uS/cm)	ASTMD1125
pH	ASTMD1293
Alkalinity (mg/L as $CaCO_3$)	ASTMD3875
TOC (mg/L)	ASTMD4839
Turbidity	ASTMD1889

filtration process. Chlorine is added to clarifier effluent line to prevent microbiological growth on the sand filter media. A static mixer installed in the clarifier effluent line for mixing the alum or chlorine solution with the clarifier effluent. Sand filter building consists of 6 continuous backwash filter cells each with 4 filter modules. Total filtration surface area is 111 sq. meters. Sand filter effluent flows into a wet well, where chlorine is added and the filtered water is discharged to the effluent storage tanks. The tertiary treated sewage effluent was analyzed by third party laboratory. Weekly analyses were conducted for BOD, COD, TSS, Total Coliform, Intestinal eggs, pH, turbidity and temperature. The monthly average concentrations of the grab samples are presented in Table 1.

2.2. Pilot system description

The pilot test was conducted with an open loop system cooling tower of 4.2 MW capacity, as shown in Fig. 3. Continuous disinfection of the recirculating water was achieved with 12.5% sodium hypochlorite solution as a primary biocide. In addition, a proprietary corrosion inhibitor (3DT Trasar 250, Nalco) was injected in the cooling tower cells for managing corrosion control during the operations. Also, a non-oxidizing biocide, Isothiozoline, was used for short period of time to control algal growth. It was discontinued as the phosphate ingredient increases the nutrient concentration. Online analyzers were used for monitoring residual chlorine, pH, and

Table 3
Bacterial analysis description.

Bacteria	LOR	Unit
klebsiella sp	1	CFU/100 mL
Legionella pneumophila SG1	10	CFU/L
Legionella pneumophila SG2-14	10	CFU/L
Legionella Species	10	CFU/L
Total Coliform	1	CFU/100 mL

temperature of the recirculating water. Chlorine residual in the range of 1.5–2.5 mg/l and pH in the range of 6.8–7.9 were observed.

2.3. Chemical water quality analysis

Reclaimed water feed line (FL) and circulating line (CL) water qualities were monitored and analyzed for water quality characteristics on a monthly basis. The parameters and corresponding analytical methods are presented in Table 2.

2.4. Bacterial analysis

Microbiological organisms, including Legionella SP, Klebsiella Pneumoniae and coliforms, were monitored for biohazard assessment with samples collected from the back end of recirculating process to reflect worst case condition. All analysis were conducted by an accredited local laboratory named ALS Arabia which maintains an A2LA accreditation of analytical testing services, which covers chemical and biological analytes under certificate numbers 3258.01 and 3258.03, 3258.04, 3258.05, 3258.09 and 3258.10. The testing laboratory complied with ISO/IEC 17025. The units and level of reporting are presented in Table 3.

In addition to water analysis, dipslide testing and biofilm swab sampling were conducted to evaluate the presence of bacteria. 21 Swab samples from internal cooling tower surfaces, sludges and structures were collected by a third-party contractor in two separate occasions: 8 swab samples during intermediate shutdown and 13 samples at the end of study shutdown and before deep cleaning. In each occasion, the samples were collected immediately after plant shut down and before deep cleaning was conducted. The locations included the basin trough, internal packing and drift eliminators.

2.5. Virus analysis

Virus analysis was conducted every two weeks at three locations of the pilot system.

- Location A: Sample was collected from the Treated Sewage Effluent (TSE) discharge line of Community Irrigation Pumps (CIP) located at the Sewage Treatment Plant. The CIP pressurizes the TSE community irrigation network which was tied into for supplying the make-up water to the cooling tower. Prior to CIP, the wastewater undergoes tertiary treatment by activated sludge (extended aeration) process, as mentioned in the method section.
- Location B: Sample was collected from the flushing tap connected to the TSE feed line for the cooling tower make up water. The sample collected from this location represents any water quality variations that may occur in the TSE while transiting from the Sewage Treatment Plant through community irrigation network during low water demand periods (water stagnation), microbiological growth and bio-film formation in the network pipes.
- Location C: Sample was collected from the sampling line connected at the back-end of the pipeline re-circulating through the cooling tower process. The sample represents the TSE water quality from the cycle of concentrations the make water-up has gone through and the condenser.

Samples were transported to King Faisal Specialized Hospital & Research Center, Riyadh in special containers, under ice within a maximum 6 h transportation time. Samples were processed under controlled conditions within 12 h of arrival. Raw Effluent/Sewage water was designed as a control for overall sample quality and integrity to recover viruses. Two molecular platforms, such as Multiplex PCR platform that is a new technology introduced in 2018 and the hybridization assay, were used in this study. KFSHRC, is an institute that holds the College American Pathology (CAP) accredited laboratory for patient care, with existing experience in using the Luminex technology for gastroenteritis pathogens detection. The lab purchased whole virus stocks from the Center for Disease Control and Prevention (CDC)/CAP, USA, for each gastroenteritis virus to be utilized as internal control and for the determination of the limit of detection. In addition to the raw sewage effluent, the second control was 1 L of sterile water spiked with two different gastroenteritis virus, in different concentration to create an internal control to validate the viral recovery results.

3. Results and discussion

3.1. Chemical water quality performance

Control of water quality parameters is required to keep cooling system from scaling, corrosion, and biofouling problems. Thus, an

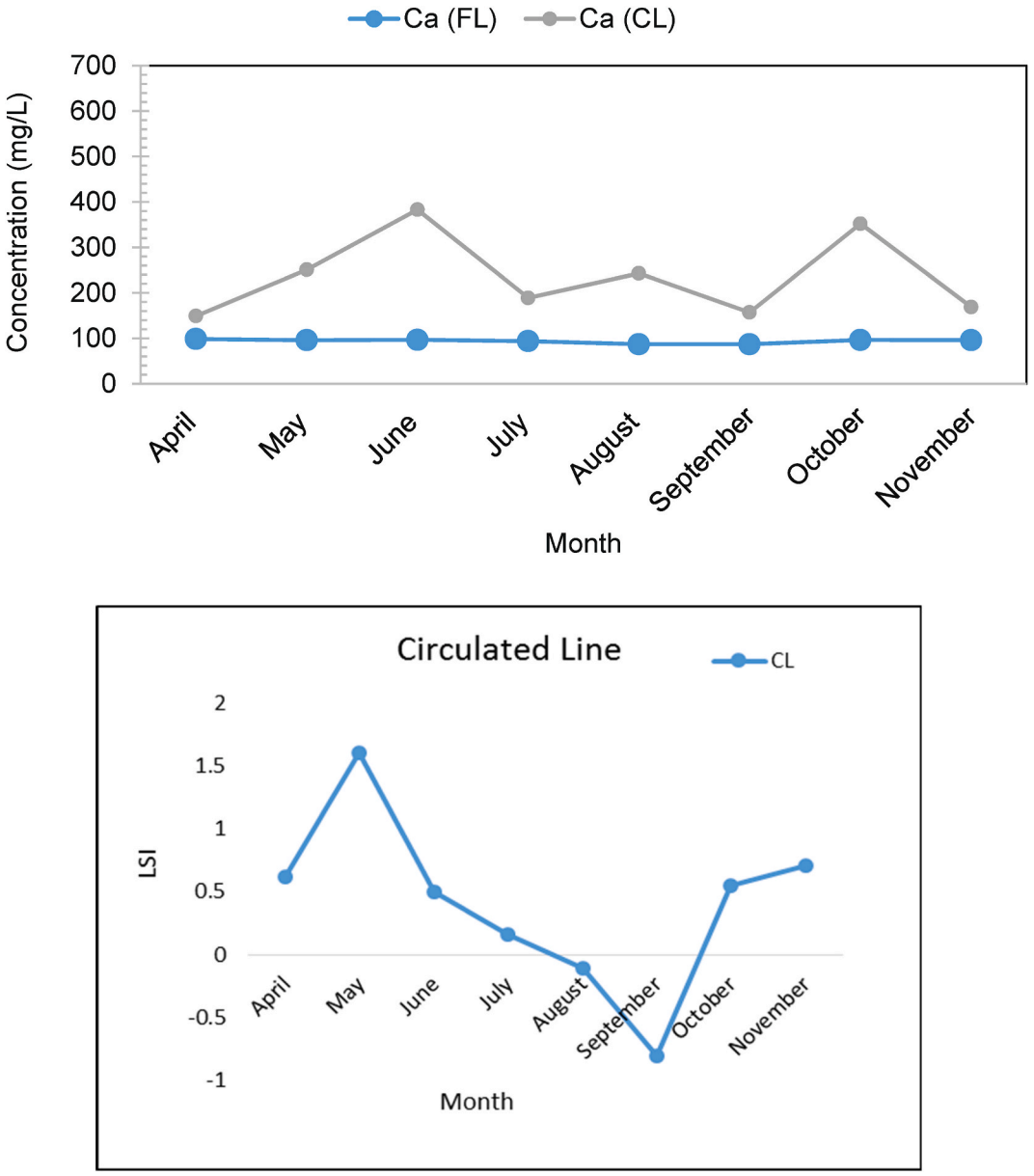


Fig. 4. Top- Calcium concentrations in the feedline (FL) and circulating line (CL) showing the accumulation of ions in the cooling system, Bottom- LSI profile in the CL line.

assessment of chemical quality characteristics is one of the principal criteria to evaluate the suitability of use of municipal reclaimed water as make-up water. Critical water quality parameters were monitored in the Feed Line (FL) and Circulating Line (CL) of the cooling system and the analysis of the data are presented below.

3.2. Scaling

Typical scale forming compounds are calcium carbonate, calcium phosphate, magnesium silicates, and calcium sulfates. These scale components adhere to the heat transfer surfaces with higher temperature because their solubilities decrease as the pH and water temperature increases. The concentrations of calcium and magnesium in the feedline remained in similar magnitude throughout the study period, but increased in the circulating line due to increase in cycle of concentration. There was an initial increase followed by a decreasing trend both in calcium ions as shown in Fig. 4 that is attributed to several trials for optimization of chemicals use concentrations.

Langelier Saturation Index (LSI) was calculated for CL as it is a measure of a solution's ability to deposit calcium carbonate. The LSI

Table 4
Metal concentrations in the flowline (makeup water) and the circulating line.

	Flow Line Concentration (mg/L)	Circulating Line Concentration (mg/L)
Manganese	<0.01	<0.01
Chromium	<0.01	<0.01
Iron	<0.01	0.09
Copper	<0.01	<0.01
Nickel	<0.01	<0.01
Lead	<0.01	<0.01
Silver	<0.01	<0.01
Aluminum	0.31	0.45
Cadmium	<0.01	<0.01
Arsenic	<0.01	<0.01
Zinc	0.03	0.05
Selenium	<0.01	<0.01

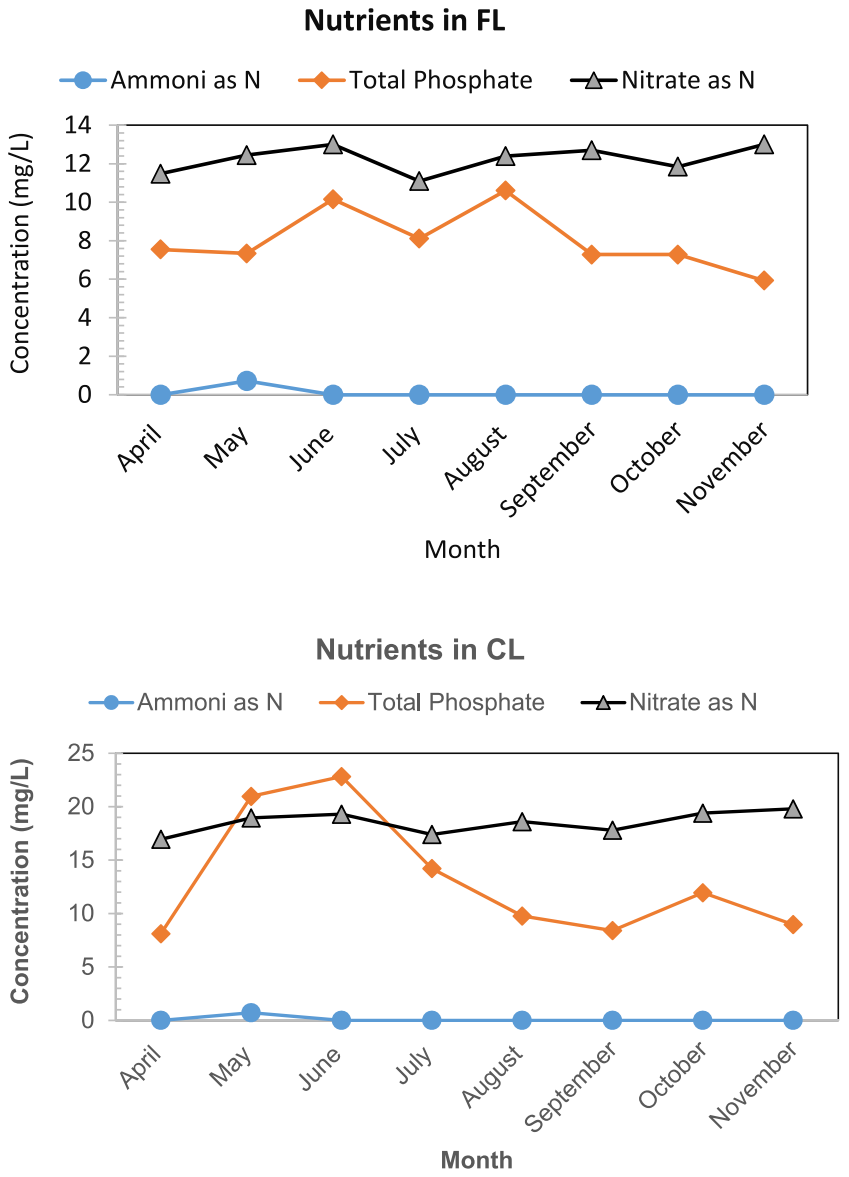


Fig. 5. Variation of nutrients in the flowline (top) and circulating line (bottom) of cooling systems.

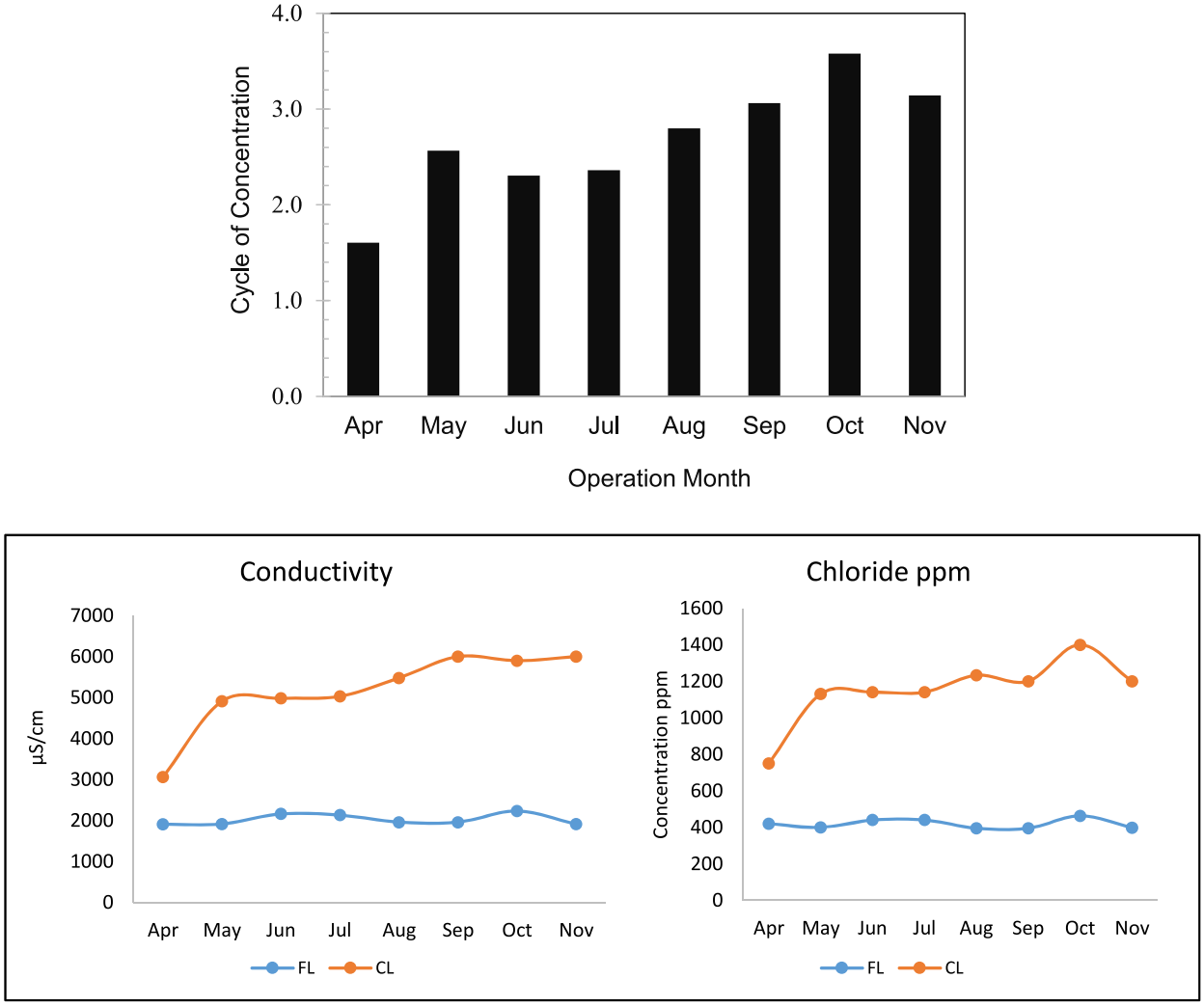


Fig. 6. Variation of cycle of concentration (top), conductivity (bottom-left), and chloride (bottom-right) in the cooling system.

values varied from 0 to 0.5 (except one data point in May), indicating that the scale forming tendency of municipal reclaimed water was low (Fig. 4). The intel temperature was 75.8 °F and the outlet temperature was 80.7 °F, with a difference across the condenser was 4.9 °F. The pH in the CL-line was 7.8 ± 0.4.

3.3. Corrosion

Corrosion is one of the concerns of the use of make-up water in cooling systems. In evaporative cooling systems, carbon steel is used for heat exchanger and piping. When the carbon steel is in contact with water, localized spots of anodes and cathodes are formed on the surface and initiate corrosion. In addition to application of a commercial corrosion inhibitor to minimize corrosion potential, metals of concern were monitored during the pilot study to evaluate the extent of corrosion, if any. The study monitored the concentration of Cr, Cu, Fe, Mn. Ni, Zn, Ag, and Al ions in both FL and CL. The geochemical analysis of water samples is shown in Table 4. The concentrations of Cu, Mn, and Ni were below the detection limits of <0.01 mg/L in FL and CL. The concentration of Fe in CL was 0.09 ppm during the study period which is low, indicating no significant corrosion. Low concentration of copper in CL is supported by low average ammonia concentrations of <0.5 ppm. However, concentration values of Zn, and Al are slightly higher compared to other ions in CL. The concentration of Zn and Al in FL was 0.03 and 0.31 mg/L respectively. Al concentration of 0.3 mg/l in FL could be attributed to residual coagulant (alum) added during secondary wastewater treatment stage. Al concentration increased to an average of 0.45 mg/L in CL due to increase in COC. From the above, it is evident that no significant corrosion occurred due to use of municipal reclaimed water in cooling system during the pilot study. The metallurgy of the cooling system includes carbon steel piping and copper tubing in the condenser. The corrosion rate of 5 mpy or less is accepted in the Company practice for this type of applications. The system did not have any corrosion monitoring program in place. However, for long-term application of TSE, the cooling systems should

undergo a regular scheduled turnaround and inspection (T&I) assessment by corrosion specialists. If the corrosion rate is found to be a concern, adequate preventive measures should be designed and implemented.

3.4. Nutrients

The nutrients present in municipal reclaimed water include NH_3, NO_3 and PO_4 ions. As microorganisms use organic matter and nutrients for their growth, high concentrations of these substances in make-up water could promote growth of microorganisms and consequent biofouling potential. The organic compounds present in the make-up water can also deteriorate the effects of biocide by consuming them or by forming the compounds of low biocidal effects. Thus, the concentrations of these nutrient ions were monitored in the FL and CL (Fig. 5). The concentration of NH_3 was <0.05 mg/L implying its removal in the wastewater treatment processes. The concentrations of NO_3 and PO_4 nutrients were high as can be seen in FL. Higher concentration of nitrates in CL is due to increase in COC. Phosphate is another nutrient that facilitates microbiological fouling and inorganic scaling. High concentration of phosphorous discharged to the environment leads to eutrophication of the receiving bodies. The average concentration of total phosphates in FL and CL was 8.0 and 13.1 mg/L respectively. It should be noted that the wastewater treatment plant was not designed for denitrification and advanced phosphate removal. However, no biological growth was noticed during the pilot study.

3.5. Water efficiency of using reclaimed water

3.5.1. Cycle of concentration

In an open circulating cooling system, dissolved salts in the make-up water are concentrated in the circulating line since the cooling water is lost through evaporation. The term COC compares the level of solids of the recirculating cooling tower to the level of solids of the original raw make up water. If the circulating water has three times the solids concentration than that of the makeup water, then the cycles are 3. A typical cooling tower is estimated to circulate about 3 L per minute per kW of cooling, and loses 1–2% to evaporation, blowdown and drift. The evaporation rate will vary depending on climate zone and cooling load/operating hours. A cooling tower discharges certain amount of water to avoid the saturation conditions and is called blowdown that includes any controlled or uncontrolled losses in the cooling tower, including drift. Thus, the COC should be controlled in order to maintain the cooling water quality and the concentration of chemicals, such as corrosion inhibitors and scale inhibitors, at their constant level by adjusting the blowdown water quantity. Blowdown is controlled by measuring conductivity of the recirculating water and varies with COC. Higher COC results in less frequent blowdown and less water use.

A simple way of determining COC is given by the ratio of conductivity in CL to conductivity in FL. In this study, conductivity and chloride concentration was used to calculate the COC as shown in Fig. 6. Prior to the pilot testing with TSE, groundwater was used for cooling tower. The groundwater characteristics consisted of a TDS level of 3040 mg/L, including scale forming ions, such as calcium (233 mg/L), magnesium (97 mg/L), sulfate (558 mg/L), and bicarbonate (226 mg/L). The COC with groundwater was limited to 2, due to higher concentration of scale forming ions. The pilot study results suggested that the COC of the cooling tower operation could be almost doubled (from 2 to 3.5) with municipal reclaimed water as make-up water as shown in Fig. 6. Higher COC means less blowdown frequency and reduction in water demand. The LSI values varied from 0 to 0.5 indicating that the scale forming tendency of municipal reclaimed water is low to moderate.

3.6. Water savings

Water savings is a function of initial makeup water volume and the change in the cycle of concentration. The data was collected for total makeup water, blow-down water, and the observed COC for the period of pilot operations. In order to obtain a water savings estimation, it was decided to treat the data in two parts: (i) Baseline condition with reclaimed water (Average of five months of operations, from April to August during which there was no optimization of operation, and the trials started with conditions similar to the operations with Groundwater) and (ii) Optimized Operation with reclaimed water (Average of three months of operation, from September to November during which the optimal operating condition was maintained). The baseline make-up water was 5172 m^3/month (i.e., average of five months of operation) and optimized make-up water was calculated at 3797 m^3/month (i.e., average of three months of operation). Therefore, the average monthly savings on makeup water as a result of the COC increase was 1375 m^3 or 27%. It is important to note that the COC can be improved with additional treatment for demineralization and biological treatment of tertiary wastewater, that will require capital investment. This paper demonstrates that the reclaimed water can replace groundwater use even without making a significant capital investment. The study findings indicate that a case-by-case evaluation based on make-up water quality and target COC, the fit-for-purpose water treatment should be developed.

3.7. Biohazard assessment of using municipal reclaimed water

During the course of the pilot, representative water samples from the cooling tower were analyzed for bacteria and viruses, as discussed below.

3.8. Bacterial study

Cooling systems, particularly open water systems are known to provide a favorable environment for micro-organisms growth and

Table 5
Bacterial and Virus analysis results of using municipal reclaimed water in the cooling system.

Type	Sampling Locations	Description	Concentrations (During April–November)
Bacteria[a]	Backend of recirculating system	Klebsiella sp	<1 CFU/100 mL
		Legionella pneumophila SG1-14	<10 CFU/L
		Legionella pneumophila SG2-14	<10 CFU/L
		Legionella Species	<10 CFU/L
		Total Coliform	<1 CFU/100 mL
Virus[b]	Raw sewage	Enteric Virus	Yes
	WWTP Post Treatment		No
	Make-up reclaimed water		No
	Circulating Line		No

[a] Bacteria samples were analysis weekly.
[b] Virus samples were collected every two weeks.

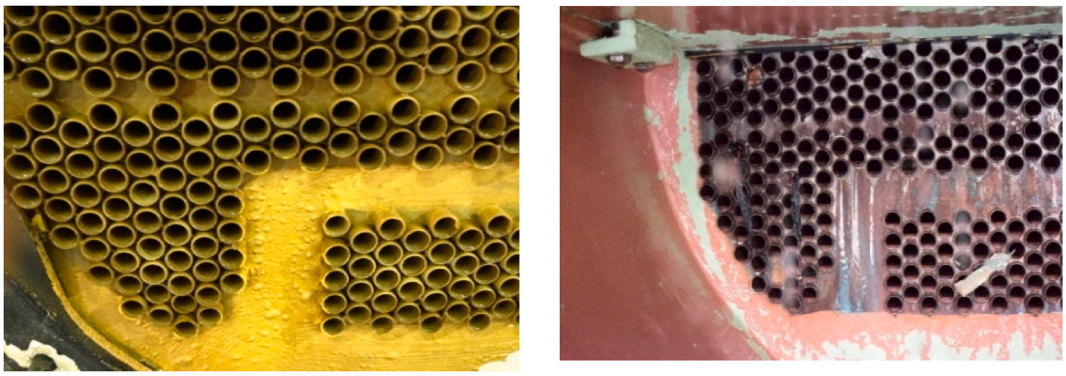


Fig. 7. Condenser surface of the cooling system operated with groundwater (Left) and with reclaimed water of this study (right)

biofilm formation. If not controlled, such biofilms can cause fouling, which can adversely affect equipment performance and promote corrosion. During the course of the pilot, representative water samples from the cooling tower were analyzed for 32 occasions for Legionella sp, Klebsiella and Total Coliforms. The results are presented in Table 5. Free residual chlorine was noted as typically operating at 2.0–2.5 mg/l and observed as minimum target levels for preventing biofilm growth; pH was typically 7.4–7.6.

The results showed that all water samples were negative for both Legionella and Klebsiella. Total coliform, an indicator for the presence of possible pathogens including all aerobic and anaerobic gram-negative bacteria, was assessed to provide an assessment of general biological contamination which is typically observed in an open structure, due to dust, vegetation and general environmental contaminants. Of the 32 samples independently tested by the laboratory, 29 of these showed coliform levels below detectable limits, indicating optimum water treatment program. The three exceptions were in: July (week 33, 750 CFU/100 mL), August (week 35, 210 CFU/100 mL) & October (week 44, 1200 CFU/100 mL). The results of these three data points were not reflected in Table 5 as they were outside of the April to November pilot duration presented in the paper. The three exceptions were likely to be due to non-optimum biocide dosing.

Legionella like to proliferate in cooling systems that are fouled with deposits and biofilm that can protect the organisms from water treatment and provide nutrients to multiply. Thus, swab samples and dipslide tests were conducted in this study. A limited number of the swab samples showed positive counts for coliform bacteria, but the absence of Legionella and Klebsiella sp. The presence is related to non-oxidizing biocide as its ingredient composition included coagulant which was causing sludge formation and deposition for a short period of time.

Dipslide tests were conducted to measure and observe microbial activity in the cooling systems. The tests conducted over 5 weeks towards the end of the pilot showed that the results were less than 2 cfu/L, indicating optimum treatment conditions in place. In addition, at the end of pilot study, additional swab and sludge residue samples were collected throughout the internal structures of cooling tower immediately after plant shut down and before deep cleaning was conducted. The locations included the basin trough, internal packing and drift eliminators. All samples were negative, demonstrating the 'cleanliness' of the system and overall efficacy of the disinfection process.

3.9. Virus study

An assessment of enteric viruses (e.g., norovirus, adenovirus, astrovirus, rotavirus, sapovirus, etc.) was also conducted during the course of pilot study. Enteric virus analysis was conducted bi-weekly during the 8-month period. The results are presented in Table 5. All the raw sewage samples contained enteric viruses. None of the samples collected from the cooling tower system operation indicated

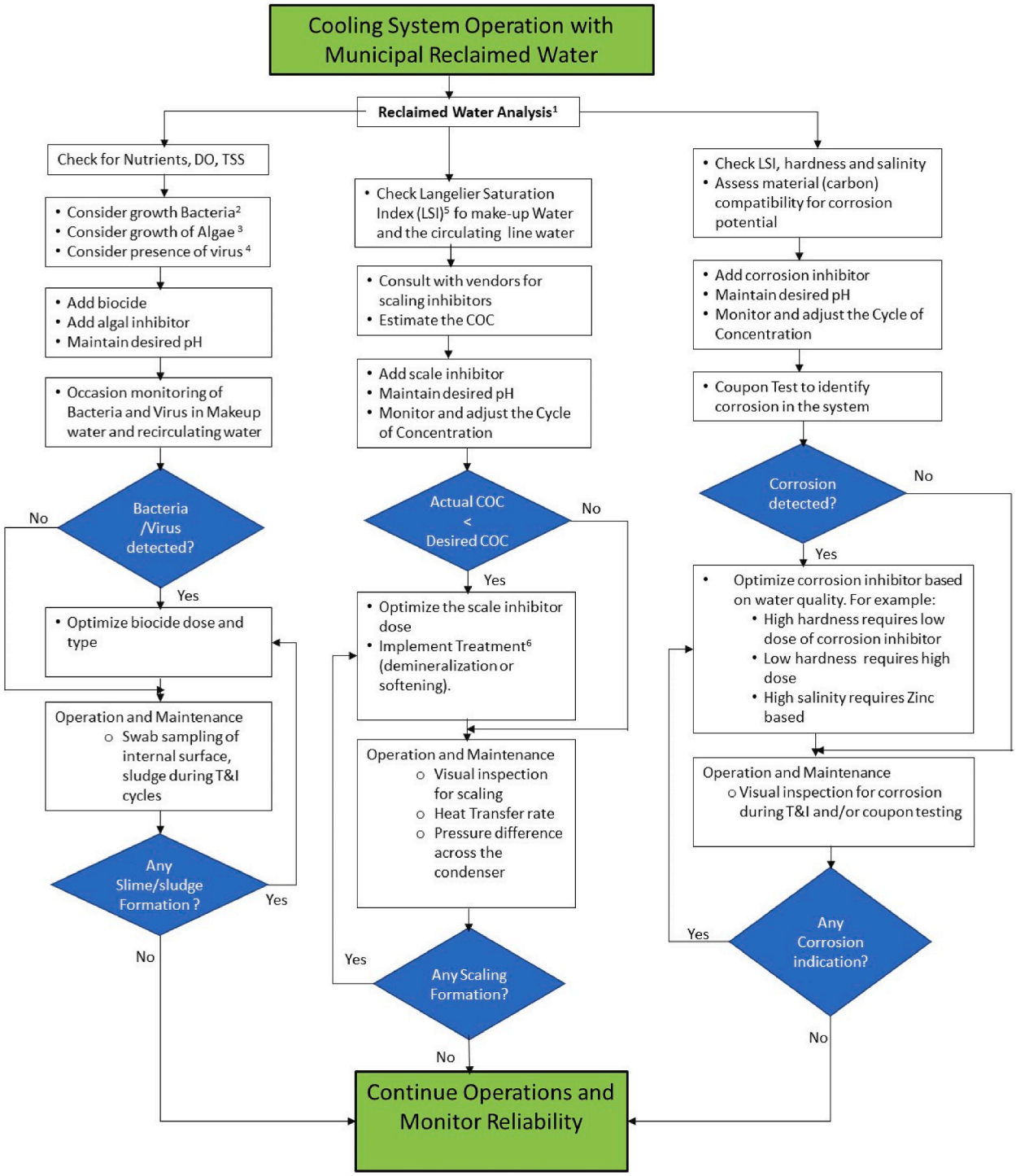


Fig. 8. Conceptual decision matrix on the water treatment optimization for the application of reclaimed water in cooling systems.

[1] *Fluctuations in reclaimed water composition may occur depending on diurnal and seasonal patterns, and wastewater treatment process reliability.*

[2] *Biological growth is of concern since reclaimed waters may contain appreciable amounts of nutrients and residual organic matter.*

[3] *Algae may grow in presence of sunlight.*

[4] *Enteric Virus should be monitored.*

[5] The Langelier Index is defined as the difference between actual pH (measured) and calculated pHs (when water saturated with calcium carbonate). The magnitude and sign of the LSI value show water's tendency to form, dissolve scale or be in equilibrium with calcium carbonate.

[6] *Treatment goals and limitations, and reclaimed water chemistry should be adequately evaluated for the selection of the best treatment technologies and process trains.*

presence of enteric viruses. The results demonstrated that the disinfection and biocide treatment was effective in controlling risks to human health. The risk from aerosols infected with viruses arising from a cooling tower system was not observed in this study and was also not readily borne out by published research on outbreaks. Furthermore, viruses will not replicate outside of the host and their environmental fate will be dependent on a range of environmental factors including temperature, humidity and sunlight.

3.10. Equipment performance and reliability of using reclaimed water

Prior to using reclaimed water, the cooling system was operated using groundwater consisted of a TDS level of 3040 mg/L. Cooling system's condenser operated with groundwater were visually inspected. As shown in Fig. 7, severe scaling was observed on the condenser surfaces with groundwater water as makeup. On the contrary, the condenser surface was clean without mineral deposit formation when municipal reclaimed water was used as make-up water. The visual comparison, as presented in Fig. 7 clearly showed the potential operational benefits of utilizing municipal reclaimed water as makeup water. It should also be noted that for ground water application, the average monthly pressure differential varied form 35–70 kPa, but for reclaimed water application, the monthly pressure difference was significantly lower and varied from 6 to 27 psi. This data also indicates that there is less potential of scaling formation in the condenser tubes while using reclaimed, which will improve the heat transfer efficiency.

4. Economic benefits

Application of the municipal reclaimed water for cooling tower operations resulted in significant savings of water, energy, and operating costs. During the pilot study, the increase in COC resulted in a 27% reduction in water demand. Because of successful outcome of the pilot study, a decision was made to utilize reclaimed water for five cooling towers of the facility with a total cooling capacity of 21 MW. For the five cooling towers, the annual expected makeup water savings for the entire facility was 82,500 m^3 groundwater. The saved makeup water also contributed to avoid energy consumption that was required for pumping and treatment of the groundwater. Annual expected energy savings from five AC plants was estimated to 247,000 KWh. During the pilot study the maintenance cost also dropped significantly when using reclaimed water. The annual maintenance costs during with groundwater utilization was $80,900, while after converting to municipal reclaimed water the maintenance cost was $15,300, which was significantly lower. Similar to the outcomes of this study, many other studies demonstrated the tangible benefits of using treated municipal reclaimed water for using as cooling system make-up water [18]. [21,22]. The pilot also shows that good make-up water quality, including acceptable TDS and low nutrient levels, is critical for cooling systems operation.

5. Conclusions

The outcome of this pilot study indicated that municipal reclaimed water is a viable water source as cooling tower makeup water in the dry hot climate. The inorganic scale forming tendency of municipal reclaimed water was lower at optimized cycle of concentration with no significant scale deposition on cooling system surfaces. The bio-hazard assessment for typical microbiological parameters associated with cooling tower operations showed that the application of a comprehensive water treatment system (disinfection), and associated robust operating procedures were effective in controlling risks to human health. In addition, the enteric virus study results demonstrated that the sodium hypochlorite treatment for inactivation of enteric viruses was effective and produced acceptable results.

In order to develop a successful reclaimed water application program for cooling systems, a number of factors need to be evaluated. The factors are: regulatory responsibilities, water quality of the reclaimed water, design and operation of the wastewater facilities generating the reclaimed water, diurnal and seasonal fluctuation in water quality, biological growth control treatment, process control and monitoring of chemical feed system, economics of using reclaimed water, and integrated planning between wastewater treatment plant and cooling tower operations. The water treatment program, including disinfection, should be designed by a qualified specialist who understands the relationship between water chemistry, microbiology, equipment material, and cooling system operations. A conceptual decision matrix on the water treatment optimization for the application of reclaimed water in cooling systems is presented in Fig. 8.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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